

Tianchu Zeng

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PROFESSIONAL SUMMARY

PhD candidate in Electrical & Computer Engineering at the National University of Singapore working in computational neuroscience and AI for science, with a focus on biomedical research. Develops AI and statistical methods to accelerate biophysical brain-model fitting (first-author paper and an invited Research Briefing at Nature Methods), improve the statistical validity of AI model evaluation, and study brain foundation models. Experienced in leading interdisciplinary projects from method design and large-scale experiments to reproducible workflows and open-source software.

RESEARCH INTERESTS

computational neuroscience · AI for science · statistical methods for AI/ML evaluation · biophysical brain models · brain imaging · brain and scientific foundation models

EDUCATION

National University of Singapore — Singapore

Aug 2022 - Present (expected 2027)

Ph.D. in Electrical & Computer Engineering

- Advisor: Assoc Prof. BT Thomas Yeo
- Co-advisor: Prof. Thomas E. Nichols, University of Oxford
- Thesis: AI for Brain Imaging
- GPA: 4.88 / 5.0
- Graduate coursework: Pattern Recognition, Visual Computing, Deep Learning, and Information Theory (all A/A-)

Tsinghua University — Beijing, China

Aug 2018 - Jun 2022

B.Eng. in Department of Automation

- GPA: 3.74 / 4.00
- Thesis: Research on Trajectory Prediction Methods for Traffic Agents in Mixed Traffic Environments
- Advisor: Prof. Jianming Hu & Prof. Yi Zhang

RESEARCH EXPERIENCE

AI for Brain Imaging — National University of Singapore

Aug 2022 - Present

- Developed DELSSOME, a deep-learning surrogate that accelerates biophysical brain-model fitting by 50–100× (and the underlying realism judgement by 1500–8000×) with parameter estimates comparable to numerical integration; first-author work accepted in principle at Nature Methods, with an accompanying invited Research Briefing also accepted.
- Led the end-to-end modeling and validation workflow and coordinated its extension to 12,005 participants across 14 datasets, yielding normative lifespan trajectories of cortical excitation-inhibition balance.
- Developed SHARP, a statistically valid test for comparing AI models under repeated cross-validation, and released a scikit-learn-compatible implementation; preprint under review.
- Led a PRISMA-guided systematic review that screened and analyzed thousands of candidate articles in journals with impact factor ≥ 15 , finding that 97% used tests that ignore dependence between cross-validation folds.
- Contributed to benchmarking brain foundation models against classical approaches for phenotypic prediction, including model fine-tuning and code review, and helped shape the lab's foundation-model research direction.
- Earlier in the PhD, developed group- and individual-level brain parcellation methods from resting-state fMRI, with functional-connectivity analysis and cortical-surface visualization in MATLAB and FreeSurfer/Freeview.

Trajectory Prediction with Graph Neural Networks — Tsinghua University

Jan 2020 - Jun 2022

- Built RNN-, GNN-, CVAE-, and Transformer-based models for multi-agent trajectory prediction and constructed a simulated Shougang Industrial Park dataset for the Beijing Winter Olympics smart-venue project.
- Work resulted in two peer-reviewed conference papers and two Chinese patent applications.

DNA Encryption/Decryption System — Tsinghua University

Sep 2020 - Jun 2021

- Led an iGEM project developing a DNA encryption/decryption system with heuristic decoding for binary data encoded into base sequences; the competition was subsequently cancelled. Advisors: Assoc Prof. Zhen Xie & Assoc Prof. Jin Gu.

Fatigue-Driving EEG Analysis — Tsinghua University

Mar 2021 - Jun 2021

- Processed driving-scenario EEG data and designed and tuned a Wide & Deep network for driver-fatigue classification.

SELECTED WORKS

* denotes equal contribution/co-first author; + denotes co-corresponding author; author name in bold.

Selected Publications

1. **Zeng, T.***, Tian, F.*, Zhang, S., Li, X., Tan, A. P., Larsen, B., ... & Yeo, B. T. (2026). Optimizing biophysical large-scale brain circuit models with deep neural networks. *Nature Methods*. Accepted in principle.
2. **Zeng, T.**, & Yeo, B. T. T. (2026). Deep learning charts the excitation–inhibition balance of the brain throughout the lifespan. *Nature Methods*. Research Briefing (invited), accompanying [1]. Accepted.
3. Qu, Y. L., Chen, J., Tam, A., Ooi, L. Q. R., Dhamala, E., Cocuzza, C. V., Zhang, S., **Zeng, T.**, ... & Holmes, A. J. (2025). Distinct brain network features predict internalizing and externalizing traits in children, adolescents and adults. *Nature Mental Health*, 3(3), 306–317.
4. Zhang, S.*, Larsen, B.*, Sydnor, V. J.*, **Zeng, T.**, An, L., Yan, X., ... & Yeo, B. T. (2024). In vivo whole-cortex marker of excitation-inhibition ratio indexes cortical maturation and cognitive ability in youth. *Proceedings of the National Academy of Sciences*, 121(23), e2318641121.

Selected Preprints

1. **Zeng, T.***, Li, H.*; Zhang, S.*; Tan, Y.Q., Tian, F., Orban, C., ..., Nichols, T.+ & Yeo, B. T.+ (2026). Widespread use of invalid statistical tests in biomedical machine learning. [bioRxiv](#) preprint; under review.

Also co-first authored two peer-reviewed transportation conference papers (CICTP 2022; ICTD 2022).

SELECTED SOFTWARE & OPEN-SOURCE PROJECTS

[admissions-ops](#) — Creator and Maintainer

- Designed and open-sourced a local-first AI-agent workspace that turns official program pages and applicant evidence into sourced evaluations, eligibility matrices, tailored CVs/resumes, and review-required statement drafts.
- Architected an "LLM authors, deterministic layer guards" workflow using reusable agent skills for Codex and Claude Code, structured file contracts, provenance tracking, and 13 CI validation targets.

[career-ops](#) — Contributor

- Contributed a merged pull request to this open-source AI job-search agent workspace (60k+ stars on GitHub).

INDUSTRY EXPERIENCE

R&D Engineer Intern — Beijing Zhangshangxianji Network Technology Co., Ltd.

Jun 2021 - Sep 2021

- Implemented Spring Boot transaction services for suppliers, retailers, and agencies.
- Improved system performance through unified Single Sign-On; built SMS authentication, OCR-resistant verification images, complex database access, and department-wide alerting with messaging and automated calls.

AWARDS, PRESENTATIONS & TEACHING

- Best Poster Award (Basic Science), NUHS Scientific and Innovation Summit, 2026.
- NUS Research Scholarship, 2022-2026.
- Invited talk at AP-CCN (Asia-Pacific Computational and Cognitive Neuroscience Conference), 2025.
- Posters at OHBM (Organization for Human Brain Mapping) 2026, NUHS Scientific and Innovation Summit 2026, and AI4X 2025.
- Cold Spring Harbor Asia Summer School on Computational and Cognitive Neuroscience, 2023.
- Teaching Assistant for EE2026 Digital Design and EE3731C Signal Processing at NUS.

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, TypeScript/JavaScript, C, C++, Java
- **AI & Scientific Computing:** PyTorch, JAX, scikit-learn, SciPy; deep learning, surrogate modeling, numerical optimization, simulation
- **Statistical Methods:** Cross-validation inference, hypothesis testing, simulation-based validation, false-positive-rate/power/confidence-interval calibration, systematic review and meta-analysis
- **Neuroimaging Analysis:** Resting-state fMRI analysis, functional-connectivity analysis, brain parcellation, cortical-surface visualization; FreeSurfer/Freeview
- **AI Agent Systems:** Codex, Claude Code; agent workflow design, tool orchestration, structured outputs, provenance tracking, deterministic validation, human-in-the-loop review
- **Software & Web:** Node.js, React, Phaser, Vite, Zustand, Capacitor
- **Research Infrastructure:** Git, GitHub Actions, Linux/bash, HPC/PBS, conda, Docker
- **Hardware & Digital Design:** FPGA (Verilog, VHDL)
- **Languages:** Chinese (native); English (TOEFL 106, GRE 326 + 4.0)